

K LAKSHMAN SIMHA

lakshmansimhakuppula@gmail.com | +91-7995825299 | Bengaluru, India | [linkedin.com/in/lakshman-simha-kuppula](https://www.linkedin.com/in/lakshman-simha-kuppula) | github.com/loki2434

PROFESSIONAL SUMMARY

Results-oriented 2025 B.Tech graduate with hands-on training in Cloud Computing and DevOps. Proficient in AWS cloud services (EC2, S3, VPC, Route 53, EFS, CloudWatch, CloudFront), CI/CD pipeline automation using GitHub Actions, and containerisation with Docker. Experienced in Linux, Bash scripting, YAML-based configuration, and Git-based version control workflows. Adept at building, deploying, and monitoring cloud infrastructure. Eager to contribute to DevOps and cloud engineering teams as a fresher.

TECHNICAL SKILLS

- **Cloud Platform:** Amazon Web Services (AWS) – EC2, S3, VPC, Route 53, EFS, CloudWatch, CloudFront
- **CI/CD & DevOps:** GitHub Actions, CI/CD Pipeline Design, Docker, Containerisation, DevOps Workflows
- **Infrastructure & Networking:** VPC Design, Subnets, Security Groups, NAT Gateway, Internet Gateway, DNS Management (Route 53), CDN (CloudFront)
- **Monitoring & Logging:** AWS CloudWatch (Metrics, Alarms, Log Groups), Basic Observability Practices
- **Version Control:** Git, GitHub (Branching, Pull Requests, Repository Management)
- **Scripting & Languages:** Bash, YAML, JSON, XML, Python (Basics), C, Embedded C
- **Tools:** Docker, Postman, VS Code, Cursor, Figma, Autodesk Fusion 360, Adobe Photoshop, Illustrator, After Effects, Blender, Unity 3D
- **Other Technologies:** Internet of Things (IoT), SAP CPI, OpenCV

EDUCATION

B.Tech CSE – IoT & Cyber Security incl. BCT, PSCMR CET

2021 – 2025

Result: 60%

WORK EXPERIENCE

IoT Engineer and Designer – Archover Solutions

October 2023 – May 2024

- Collaborated in a five-member team to research and develop a Patient Monitoring system using IoT sensors.
- Integrated and designed embedded hardware for a patient monitoring device, connecting multiple sensors for real-time data acquisition.

PROJECTS

AWS Multi-Tier VPC Architecture – Cloud Infrastructure

2026

- Designed and deployed a production-style 3-tier AWS architecture comprising a custom VPC with public and private subnets, an Application Load Balancer bridging subnets, and an Auto Scaling Group of EC2 instances with attached IAM roles.
- Configured Route 53 DNS and CloudFront CDN with AWS WAF in front of the ALB for secure, low-latency global content delivery.
- Provisioned a private MySQL RDS database subnet with security groups restricting access to application-tier instances only, alongside an Internet Gateway and NAT Gateway for controlled inbound/outbound routing.
- Implemented end-to-end observability using CloudWatch metrics and alarms integrated with SNS notifications and SQS queues for automated alerting.

IoT Sensor Simulator – Dockerized CI/CD Pipeline – Python, Flask, Docker, GitHub Actions, Docker Hub

2026

- Built a Flask service simulating IoT sensor data and containerized it with Docker.
- Set up a GitHub Actions pipeline that runs tests, builds the Docker image, and pushes it to Docker Hub on every push to main.

GitHub: <https://github.com/loki2434/iot-app-pipeline>

DevOps Home Lab Automation – Shell Script – Bash, Shell Scripting, Linux

- Wrote a Bash script to automate installation and configuration of 9 core DevOps tools (Docker, kubectl, Helm, Terraform, Ansible, AWS CLI, Jenkins, Prometheus, Grafana) on Ubuntu/Debian Linux.
- Implemented root privilege check, set -euo pipefail error handling, GPG key setup, third-party apt repository configuration, and systemd service activation in a single automated run.

GitHub: <https://github.com/loki2434/devops-tools>

Visitor Management System and ID-Pass Generation

Mar – Apr 2025

- Built a full-stack application to capture visitor details (name, phone, purpose, photograph via OpenCV) and auto-generate a digitally signed entry pass, improving college security and visitor tracking.

POSITIONS OF RESPONSIBILITY

Social Media Head – Cycurithings Student Chapter Club

2024 – 2025

- Managed all club social media platforms; designed posters, brochures, certificates, and ID cards; led a team of four and coordinated scheduling and promotion of college-level competitions.

ACHIEVEMENTS

- 1st Prize – IoT Domain, 24-Hour Hackathon, KPRIET (Child Safety IoT Device)
- 2nd Position – Project Expo, NRI College (out of 20+ teams)
- 3rd Prize – 'Prompted Pixel' Competition, MIC College of Engineering
- Silver Medal – Boxing (65 kg), Aboxx Fight Night, Delhi – May 2022